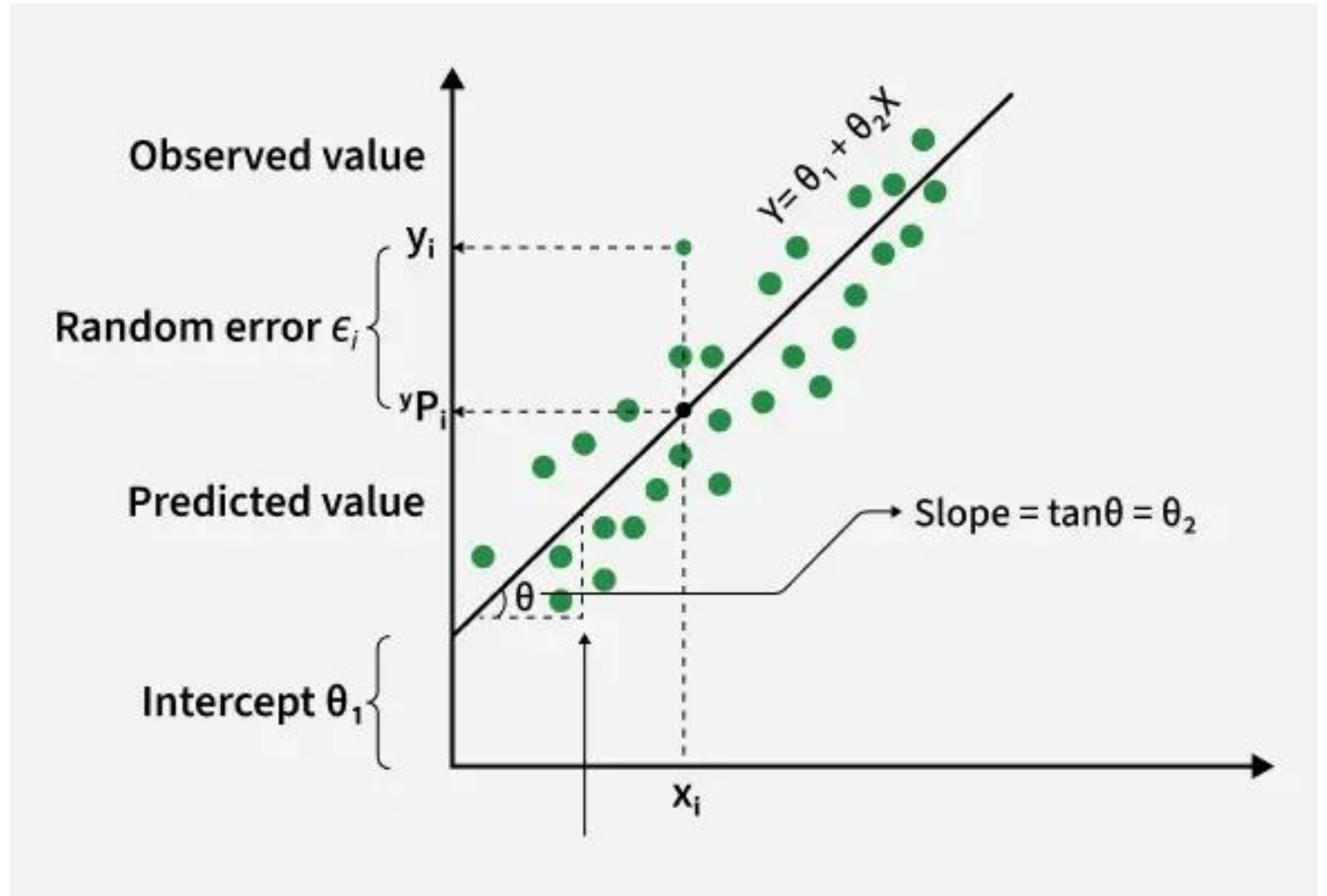


Simple Regression: Predicting Numbers

Drawing the Line of Best Fit into the Future

Instructor: Zaryab Ahmad Khan



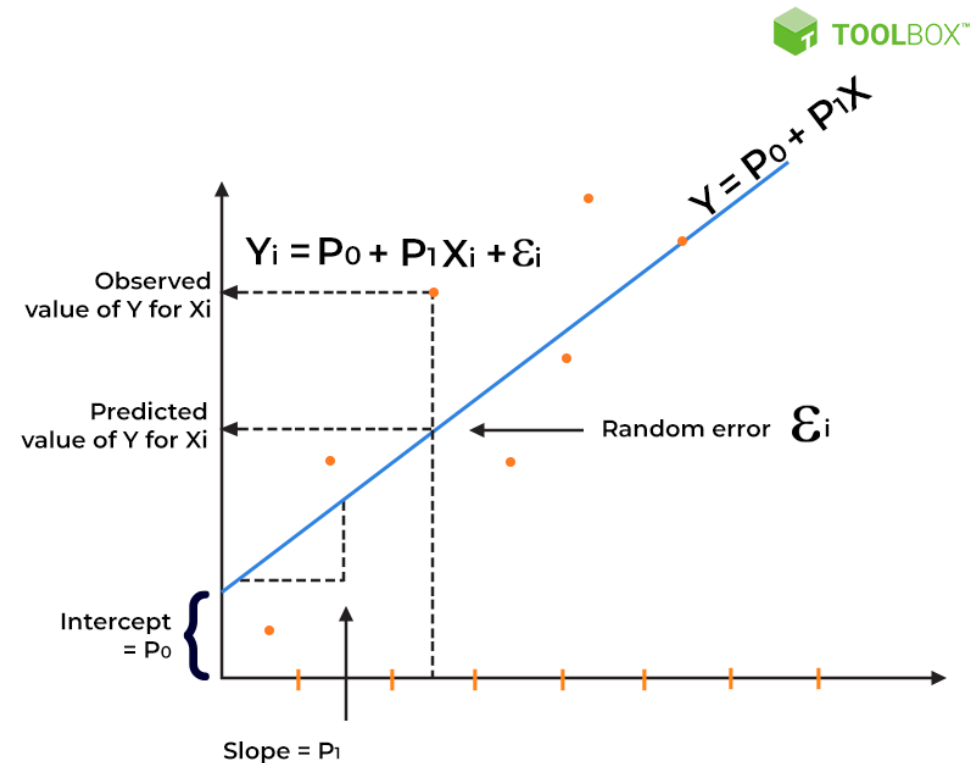
Simple Linear Regression (The Foundation)

- **Concept:** Predicting an outcome using a *single* variable.
- **How it works:** The algorithm draws the mathematical "Line of Best Fit" straight through your data points on a graph.
- **Real-World Use:** Predicting crop yield based solely on the amount of rainfall.

$$y' = b + w_1 x_1$$

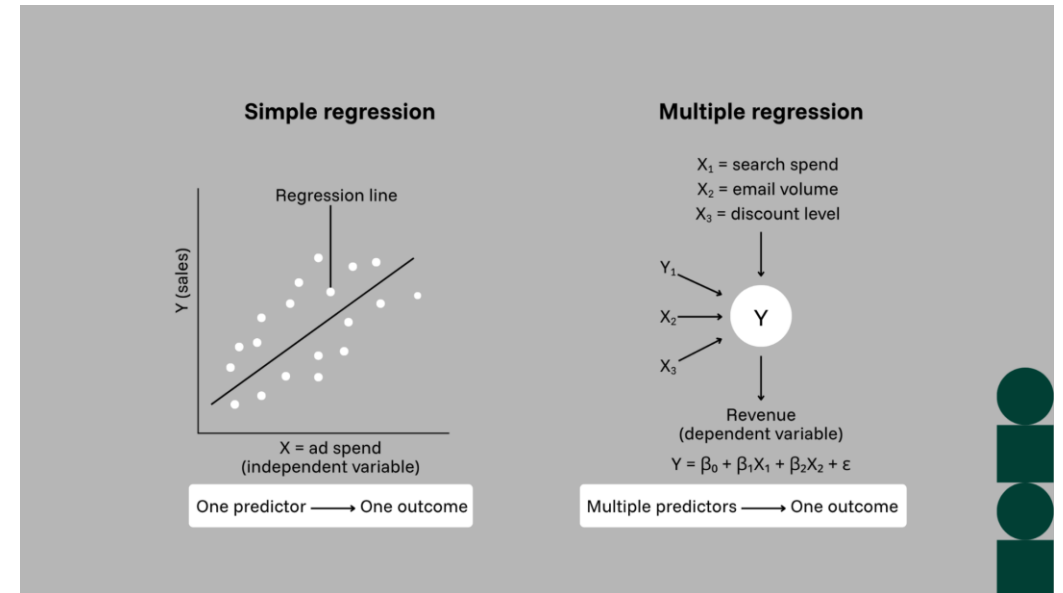
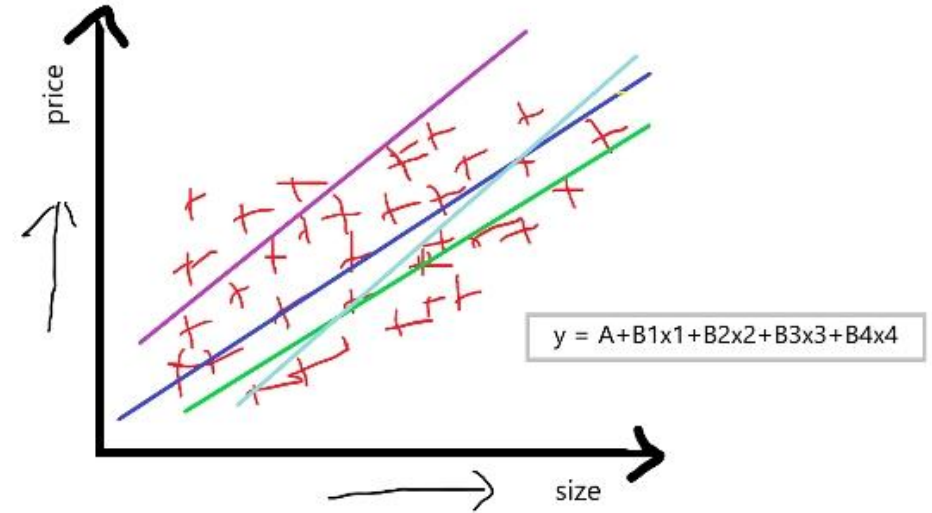
Prediction Bias Weight Feature value

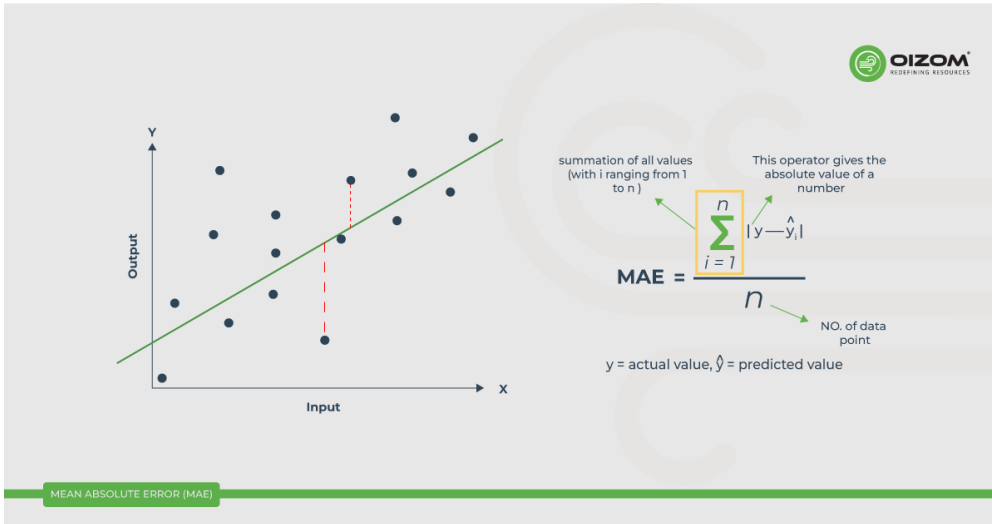
Calculated from training



Multiple Linear Regression (Adding Complexity)

- **Concept:** Predicting an outcome using *multiple* variables at once.
- **How it works:** Instead of a 2D line, the math scales up to multiple dimensions (which humans can't easily visualize, but computers handle perfectly).
- **Real-World Use:** Predicting a house price in Peshawar using size, location, *and* number of bedrooms.





Evaluating the Model (Are We Right?)

- **The Problem:** The AI will always give you a prediction, but how do we know exactly how wrong the predictions are in real numbers?
- **Metric 1: Mean Absolute Error (MAE):** The average amount our prediction was off by. If predicting house prices, an MAE of "5" means we are off by 5 Lakhs on average.
- **Metric 2: R-Squared (R²):** A score from 0 to 1 (or 0% to 100%) showing how well our features explain the target. 0.90 is an excellent "A grade" model!

$$\text{R-Square} = \frac{\sum(Y_{\text{actual}} - Y_{\text{predicted}})^2}{\sum(Y_{\text{actual}} - Y_{\text{mean}})^2}$$